

The paradox of the derivative

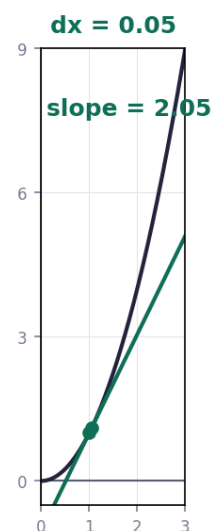
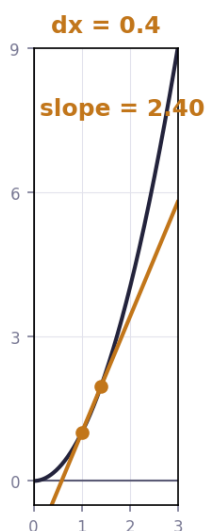
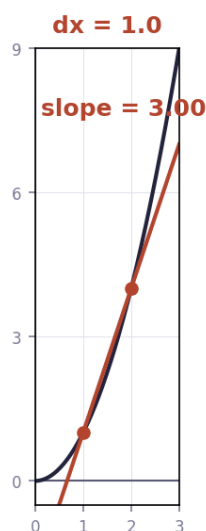
Rate of change at a single instant.

THE IDEA

A derivative measures how fast something changes *at one instant* — but change needs two points to measure. That is the paradox. The resolution: take two points a distance dx apart, compute the slope, then ask what that slope **approaches** as dx shrinks toward zero.

The definition: $df/dx = [f(x + dx) - f(x)] / dx$, as $dx \rightarrow 0$.

as dx shrinks, the slope approaches the true derivative $2x = 2$



WHAT THE DERIVATIVE TELLS YOU

Sign of df/dx	Meaning
positive	The function is increasing
negative	The function is decreasing
zero	Flat — a peak, a valley, or a plateau

Geometrically it is the **slope of the tangent line** at that point.

WORKED EXAMPLE

CHAPTER 2 — WORKED EXAMPLE

the derivative from first principles

THE DEFINITION

$$\frac{df}{dx} = [f(x + dx) - f(x)] / dx \quad \text{as } dx \rightarrow 0$$

EXAMPLE $f(x) = x^2$

$$\begin{aligned} f(x + dx) &= (x + dx)^2 \\ &= x^2 + 2x \cdot dx + (dx)^2 \end{aligned}$$

subtract $f(x) = x^2$:

$$= 2x \cdot dx + (dx)^2$$

divide by dx :

$$= 2x + dx$$

let $dx \rightarrow 0$:

$$\frac{df}{dx} = 2x$$

THE PARADOX

"rate of change at one INSTANT" — but change needs two points.

answer : it is what the slope APPROACHES as dx shrinks to nothing.

Why it matters for ML: gradient descent finds where the derivative is zero — the bottom of the loss valley. The sign tells the optimiser which way to step; the size tells it how big a step to take.

FORMULAS FROM THIS CHAPTER

Definition	$\frac{df}{dx} = [f(x+dx) - f(x)] / dx, \text{ as } dx \rightarrow 0$
$\frac{d}{dx} (x^2)$	$2x$
Positive derivative	function increasing
Zero derivative	flat — peak, valley or plateau

TEST — Chapter 2

1. For $f(x) = x^2$, compute the derivative at $x = 3$ using $df/dx = 2x$. → **6**
2. Expand $(x + dx)^2$ and identify the term that survives after dividing by dx and letting $dx \rightarrow 0$. → **$x^2 + 2x \cdot dx + (dx)^2$; the surviving term gives $2x$**
3. If $df/dx = 0$ at a point, what does that tell you about the graph there? → **it is flat — a peak, valley, or plateau**